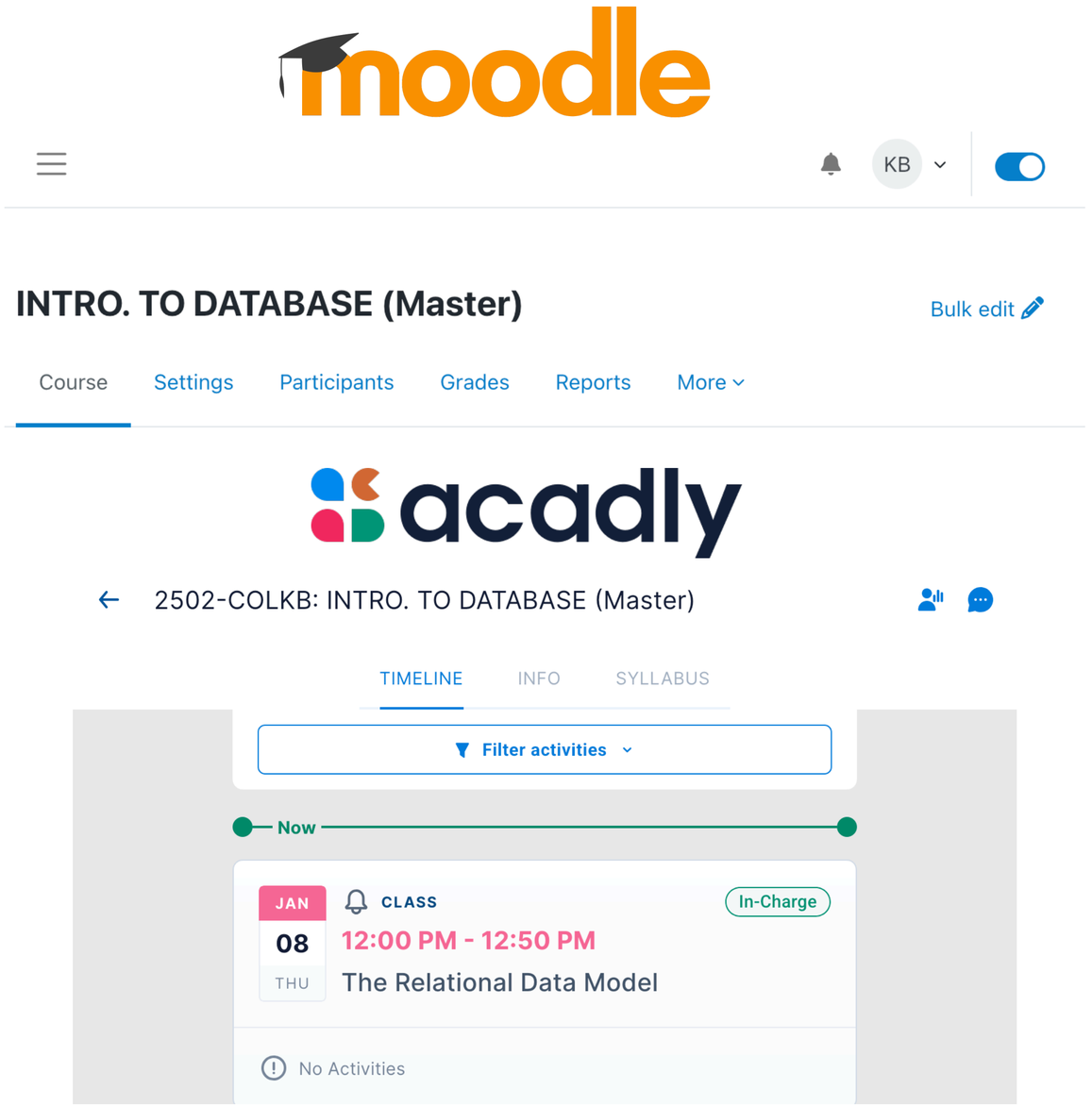
COL 362/632
Introduction To Database Management Systems

Relational Model

Kaustubh Beedkar

Administrivia

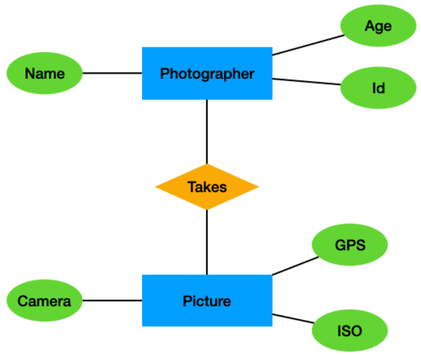
- No class on Saturday 10.01.2026!
- I'll take attendance starting next week (make sure to have access to Acadly)
- Moodle New (https://moodlenew.iitd.ac.in/course/view.php?id=9217)
- Acadly (https://app.acadly.com/iitdelhi/courses/7c08e417)
- Course Webpage (https://web.iitd.ac.in/~kbeedkar/#teaching)



Syllabus

Overview

1. Design and Programming
   - Data Modelling
     - ER Model
     - Relational Model
     - Schema Design
   - Relational Algebra
   - SQL (Basic + Advanced)
2. Implementation and Internals
3. Misc. Topics



From RealWorld to Database Schema

Database Schema

Photographer			
First Name	Last Name	Age	ID

Picture			
Camera	GPS	ISO	Apperture

Photographer = {[Name: String, Age: Integer, ID: Integer]}
Picture = {[Camera: String, GPS: Point]}

Relational Model

A Historical Anecdote

Information Management System (IMS)

IBM

Initial release	1966; 60 years ago
Stable release	15.6.x[1] / June 13, 2025; 6 months ago[1]
Operating system	z/OS at least 02.05.00 [2]
Platform	IBM System z
Type	Database & transaction processing subsystem
License	Proprietary
Website	www.ibm.com/software/data/ims/index.html

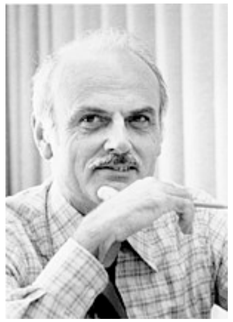
E.F.Codd Publishes

Information Retrieval — P. BAXENDALE, Editor

A Relational Model of Data for Large Shared Data Banks

E. F. Codd
IBM Research Laboratory, San Jose, California

The relational view (or model) of data described in Section 1 appears to be superior in several respects to the graph or network model [3, 4] presently in vogue for non-inferential systems. It provides a means of describing data with its natural structure only—that is, without superimposing any additional structure for machine representation purposes. Accordingly, it provides a basis for a high level data language which will yield maximal independence between programs on the one hand and machine representa-



Too Slow, too theoretical

IBM remains skeptical

WIKIPEDIA
The Free Encyclopedia

IBM System R — 12 languages

Article Talk Tools

From Wikipedia, the free encyclopedia

IBM System R is a database system built as a research project at IBM's San Jose Research Laboratory beginning in 1974,[1] led by Edgar Codd, to implement his ideas on relational databases.[2] System R was a seminal project as the first implementation of SQL, which has since become the standard relational data query language. It was also the first system to demonstrate that a relational database could provide good transaction processing performance. Design decisions in System R, as well as some fundamental algorithm choices (such as the dynamic programming algorithm used in query optimization[3]), influenced many later relational systems.

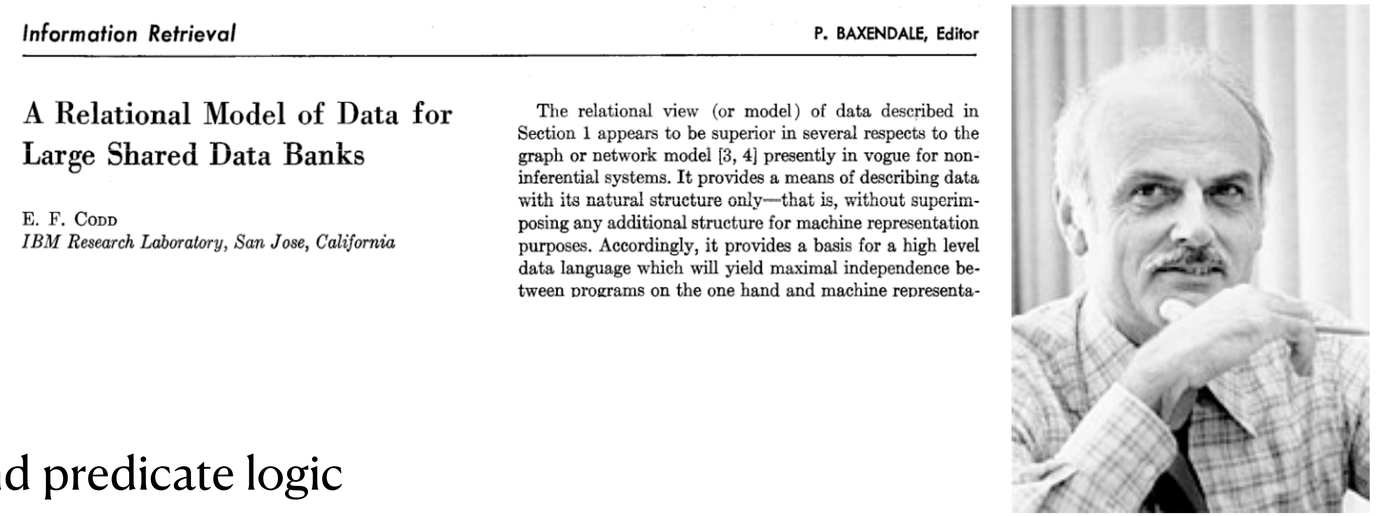
Oracle commercialises relational DBs first

ORACLE

Relational Software Systems, Inc.

1960s — 1970 — Early-1970 — Mid-1970 — 1980s

F.Codd and the Relational Model



Information Retrieval P. BAXENDALE, Editor

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Turing Award (1981)

- Proposed by Edgar Frank Codd in 1970
- Has just one concept that of **relation**
  - Mathematical concept based on set theory and predicate logic
  - — a set of tuples (rows) with defined attributes (columns)
  - — provides strong theoretical foundations
- A database table represents a relation, with rows as instances and columns as attributes
- Simple but very powerful concept
- Applicable across domains and forms the basis of modern database management systems

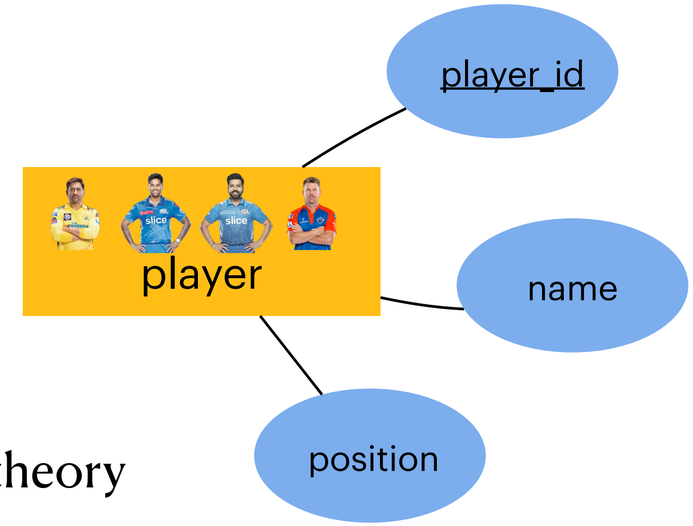
"The relational model is not just a way of organizing data; it's a way of thinking about data."

Relational Model and Its Mathematical Foundations

- $R \subseteq D_1 \times D_2 \times \ldots \times D_n$
- D_i is domain or range of values
  - Specifies the set of values that an attribute can take
- R is the expression or instance
- Example: Player ⊆ int x string x int
- **Relational Algebra:** formal query language for relational model based on set theory and predicate logic — more on this in the next lecture
- Schema of a relation = relation name + attribute name
- $[R] : \{[\text{attribute}_1 : D_1, \ldots, \text{attribute}_n : D_n]\}$
- $D_i = \text{dom}(\text{attribute}_i)$
- Example

[Player] : {[player_id:int, name:string, position:int]}

Player ⊆ dom(player_id) x dom(name) x dom(position)



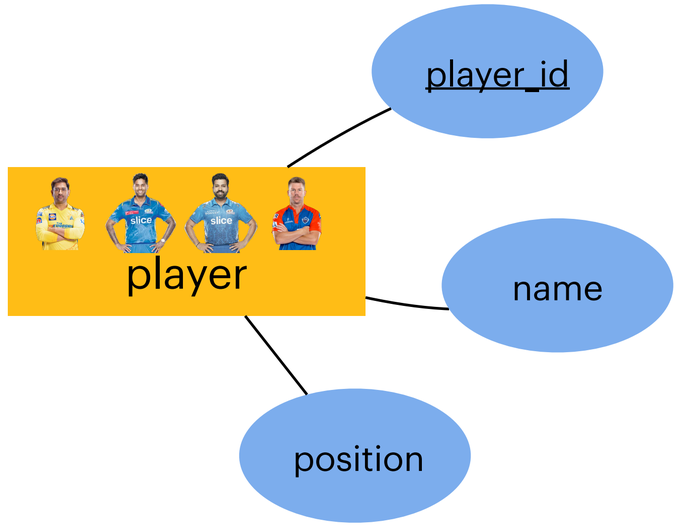
PlayerId	Name	Position
123	MS Dhoni	7
131	Rohit Sharma	1
134	SK Yadav	5
154	David Warner	3

Relational Model and Its Mathematical Foundations

Instance vs Schema

- Relation Player is an instance of the relation schema [Player]
- Instance is a set of records
- Example [Player] : {[player_id:int, name:string, position:int]}

  Player = {(123, MSD, 7), (134, SKY, 5), (131, RS, 1), (154, DW, 3)}



Tuple

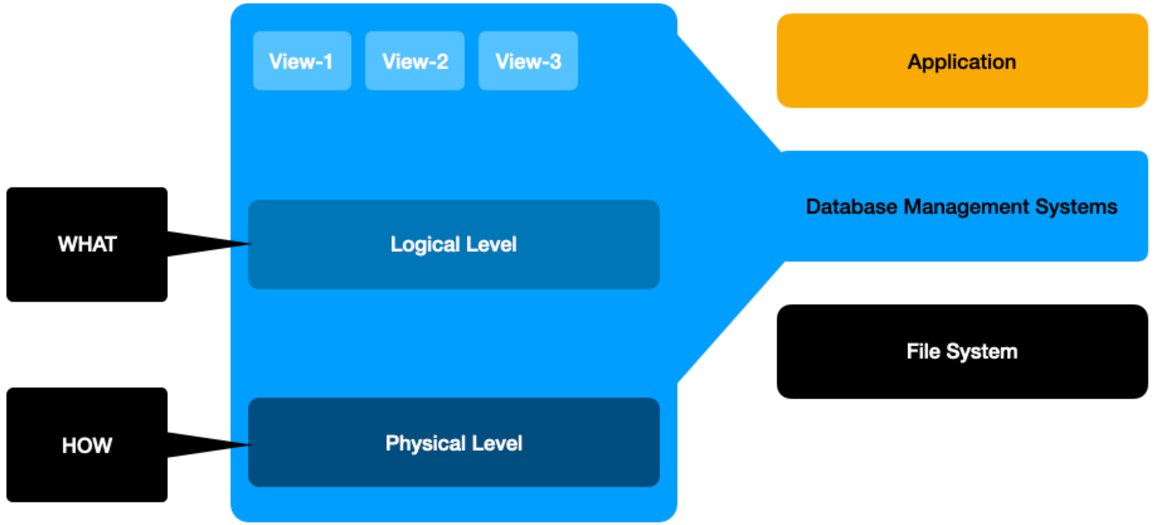
- Also known as **record**
- $t \in R$
- Example Player = {(123, MSD, 7), (134, SKY, 5), (131, RS, 1), (154, DW, 3)}
  t = (134, SKY, 5)

PlayerId	Name	Position
123	MS Dhoni	7
131	Rohit Sharma	1
134	SK Yadav	5
154	David Warner	3

Relational Model and Its Mathematical Foundations

Data Independence

- One of the most powerful ideas!
- You can change storage layouts, indexes, distribute data across machines. But your applications will never break
- This is why a database survives hardware changes/technological evolution
- Very few abstractions in Computer Science have aged this well



Relational Model and Its Mathematical Foundations

Relational Abstraction: Universal Interface for Data

- Programmers don’t think in terms of files and pointers
- Analysts don’t write code—they write *queries*
- Data becomes accessible to **non-programmers**

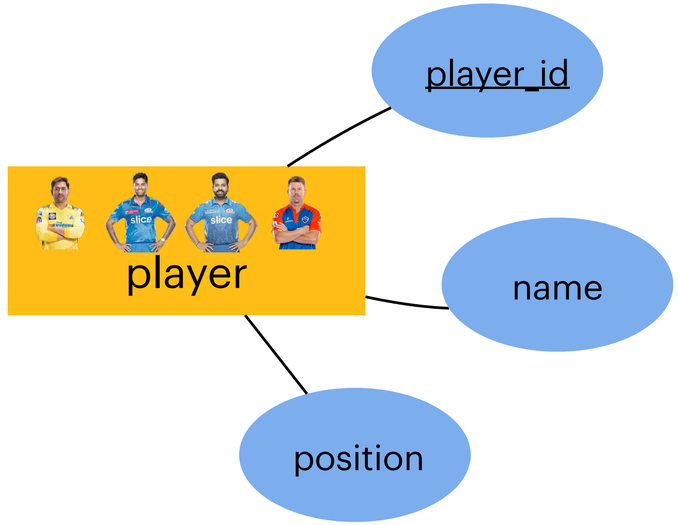
Does this sounds familiar in the LLM era?

ER Model to Relational Model

Mapping Entity set

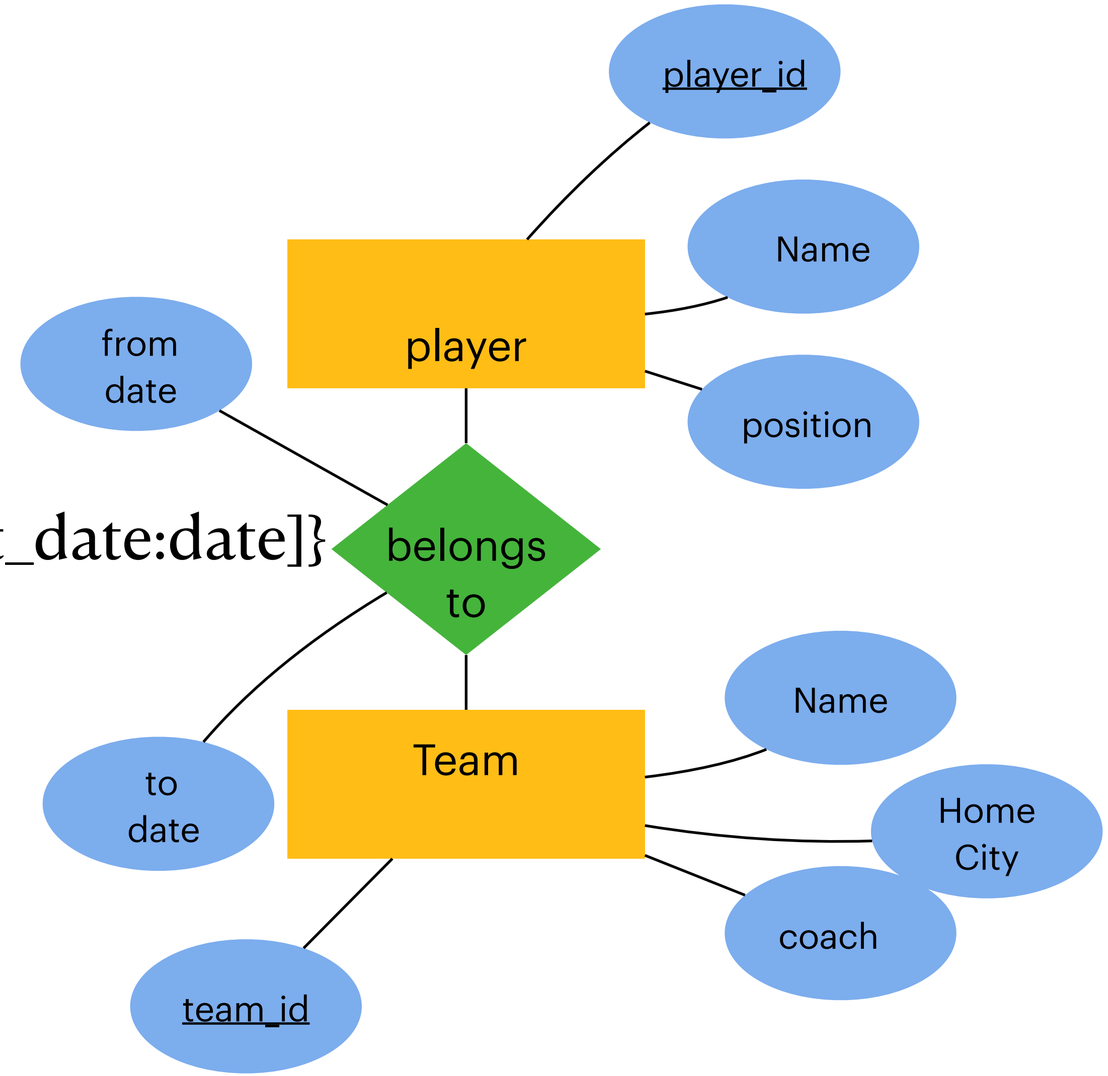
- Convert each strong entity type to a relation
- Include attributes and specify the (primary) key

- [Player]: {[player_id:int, name:string, position:int]}



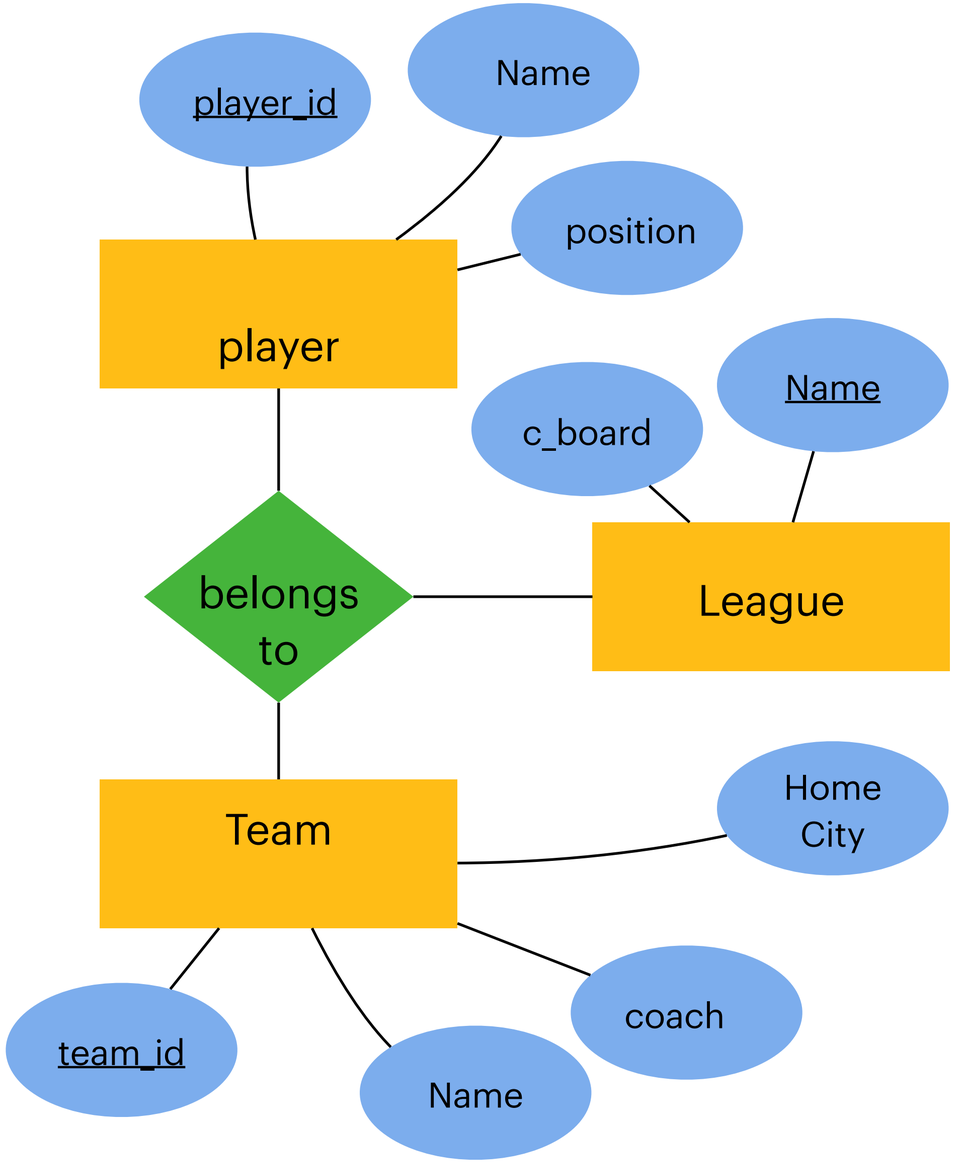
Mapping Relationship

- [Belongs]: {[player_id:int, team_id:int]}
- [Belongs]: {[player_id:int, team_id:int, f_date:date, t_date:date]}



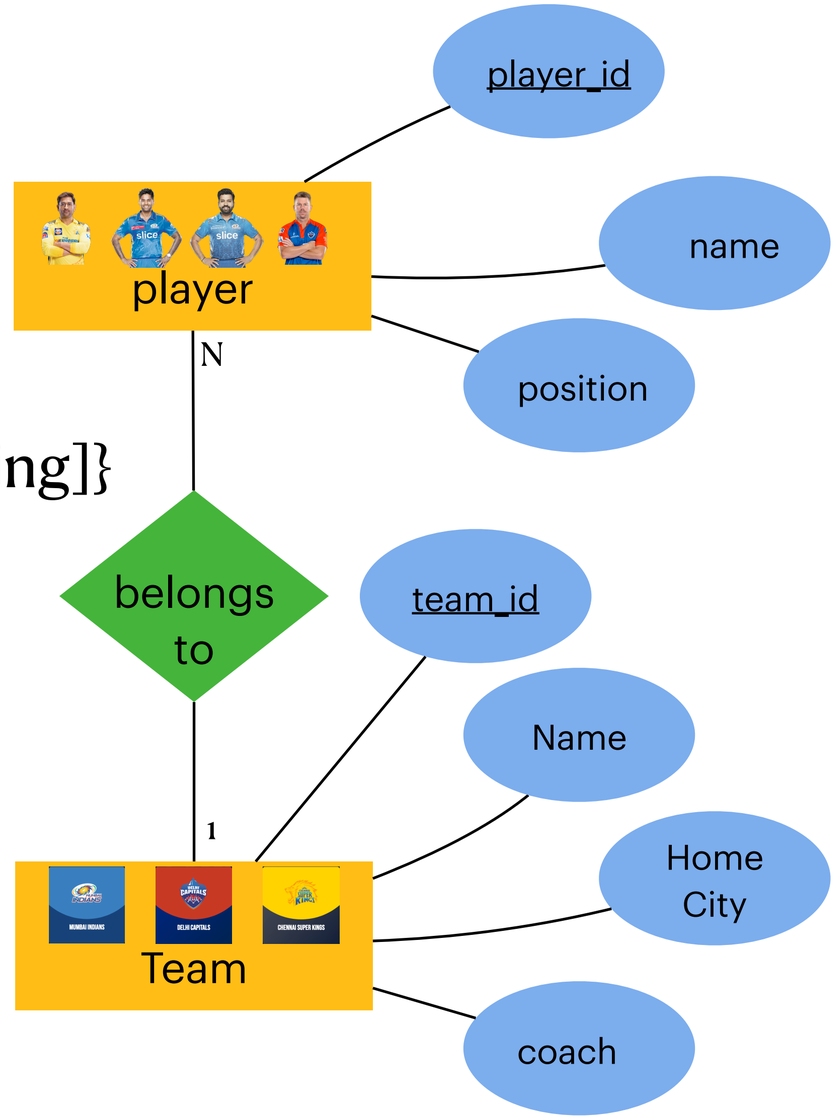
Mapping Multiway Relationship

- Create a new relation with (foreign) keys of all participating entities
- [Belongs]: {[player_id:int, team_id:int, l_name:string]}



1:N (Many-One) Relationship

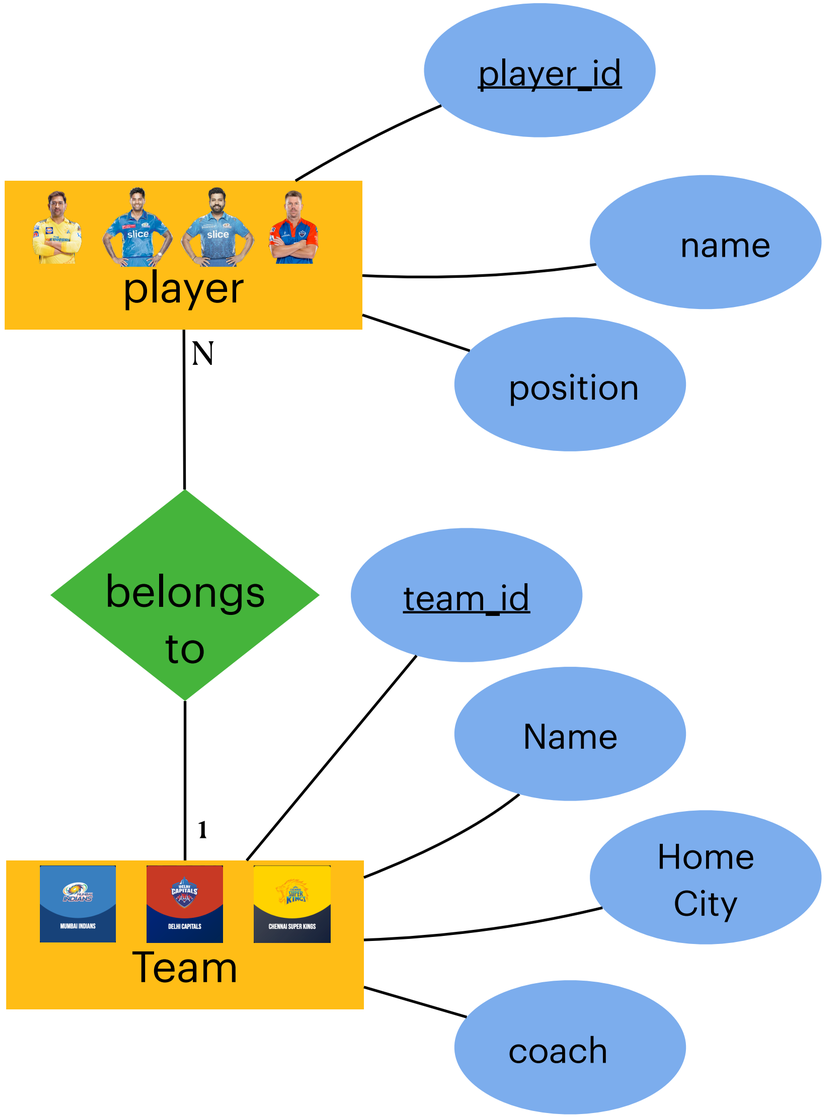
- [Player]: {[player_id:int, name:string, position:int]}
- [Belongs]: {[player_id:int, team_id:int]}
- [Team]: {[team_id:int, name:string, home:string, coach:string]}



1:N (Many-One) Relationship

- Modify the Player relation
- Add a (foreign) key to one of the participating relations

- [Player]: {[player_id:int, name:string, position:int, plays_for: int]}
- ~~[Belongs]: {[player_id:int, team_id:int]}~~
- [Team]: {[team_id:int, name:string, home:string, coach:string]}



Can we modify the Team relation instead?

Always add a (foreign) key to the relation on the “many” side

1:1 (One-One) Relationship

Add a (foreign) key to one of the participating relations

Option-1

[Player]: {[player_id:int, name:string, position:int, captain_of:int]}

[Team]: {[team_id, name:string, home:string, coach:string]}

Option-2

[Player]: {[player_id:int, name:string, position:int]}

[Team]: {[team_id:int, name:string, home:string, coach:string, captain_id:int]}

Which is better? (Option-2 is better as the option-1 will have a lot of null values)

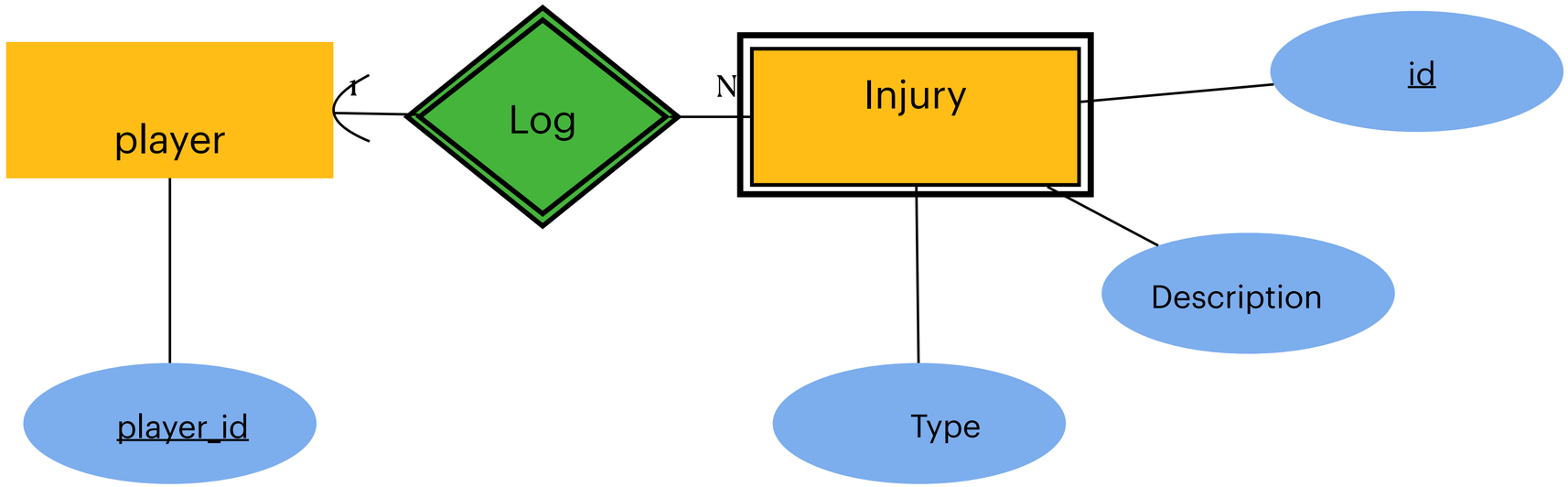
M:N (Many-Many) Relationship

- [Player]: {[player_id:int, name:string, position:int, plays_for:int, sponsored_by:int]}

Whats wrong here?

- **Always create a new relation with (foreign) keys referencing the participating entities**
- [Player]: {[player_id:int, name:string, position:int, plays_for:int]}
- [Sponsor]: {[player_id:int, company_id:int]}
- [Company]: {[company_id:int, name:string, manager_id:int]}

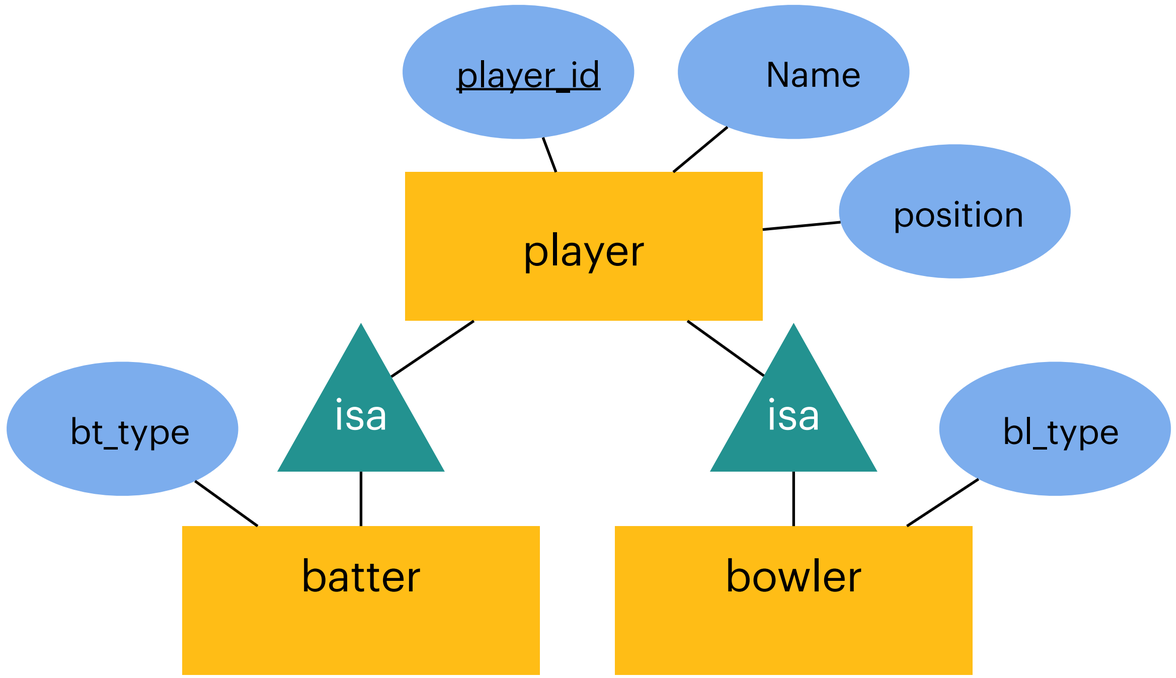
Mapping Weak Entity Sets



- Include (foreign) key referencing the related strong entity

- Combine primary key of the strong entity with weak entity discriminator
- [Injury]: {[id:int, player_id:int, type:string: description:string]}

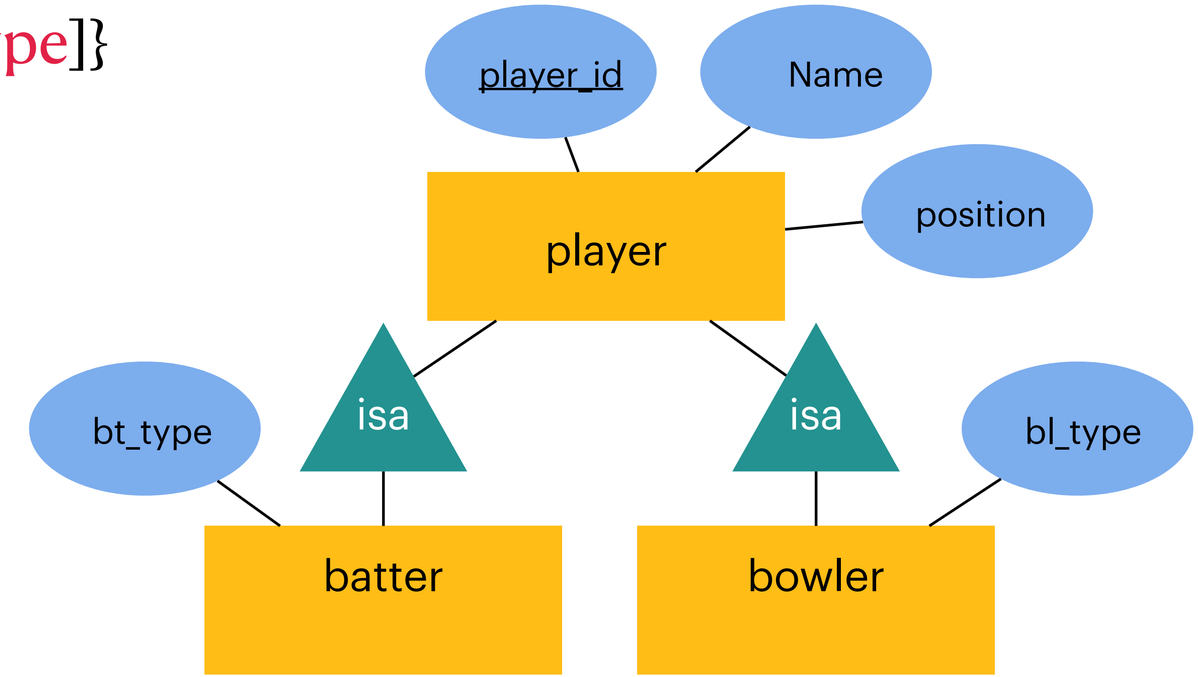
Mapping IsA Relationship



IsA Relationship

Option-1 (create one relation)

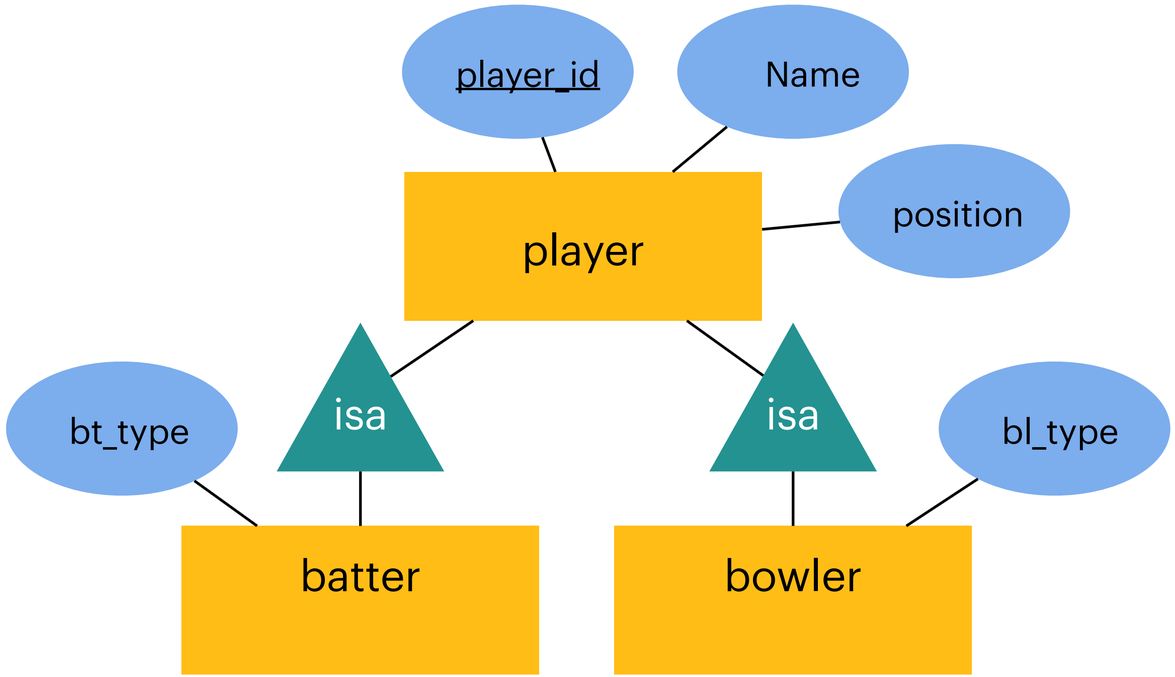
- [Player]: {[player_id, name, pos, bt_type, bl_type]}



IsA Relationship

Option-2

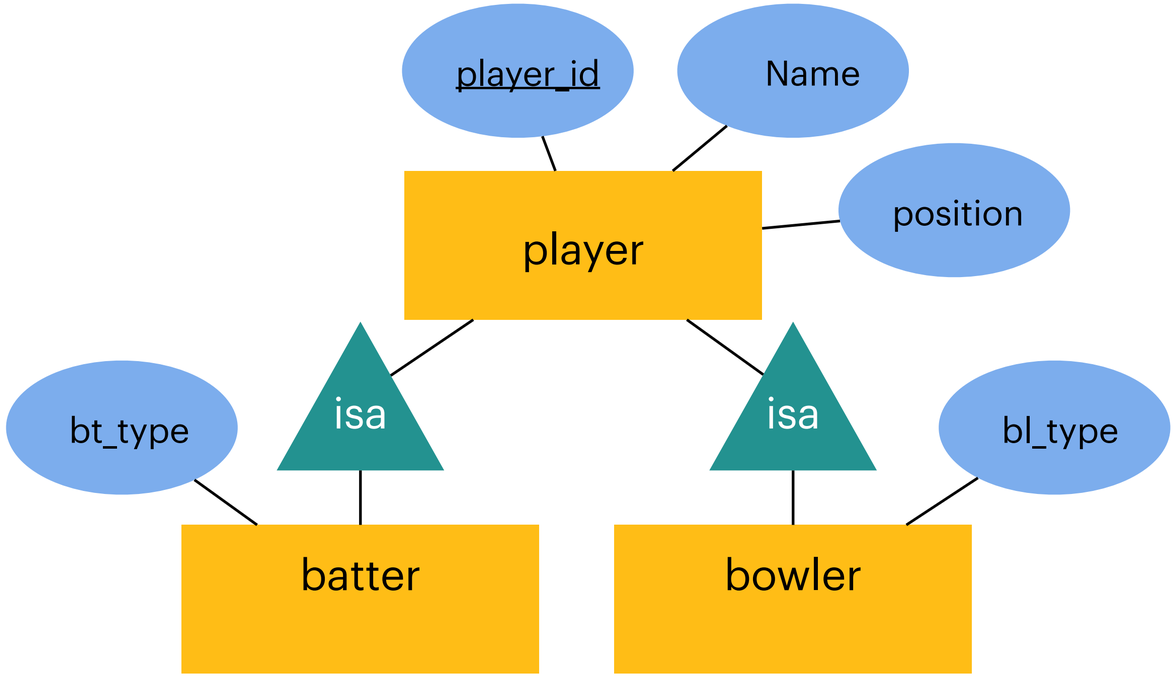
- [Player]: {[ ]}
- [Batter]: {[player_id, name, pos, bt_type]}
- [Bowler]: {[player_id, name, pos, bl_type]}



IsA Relationship

Option-3

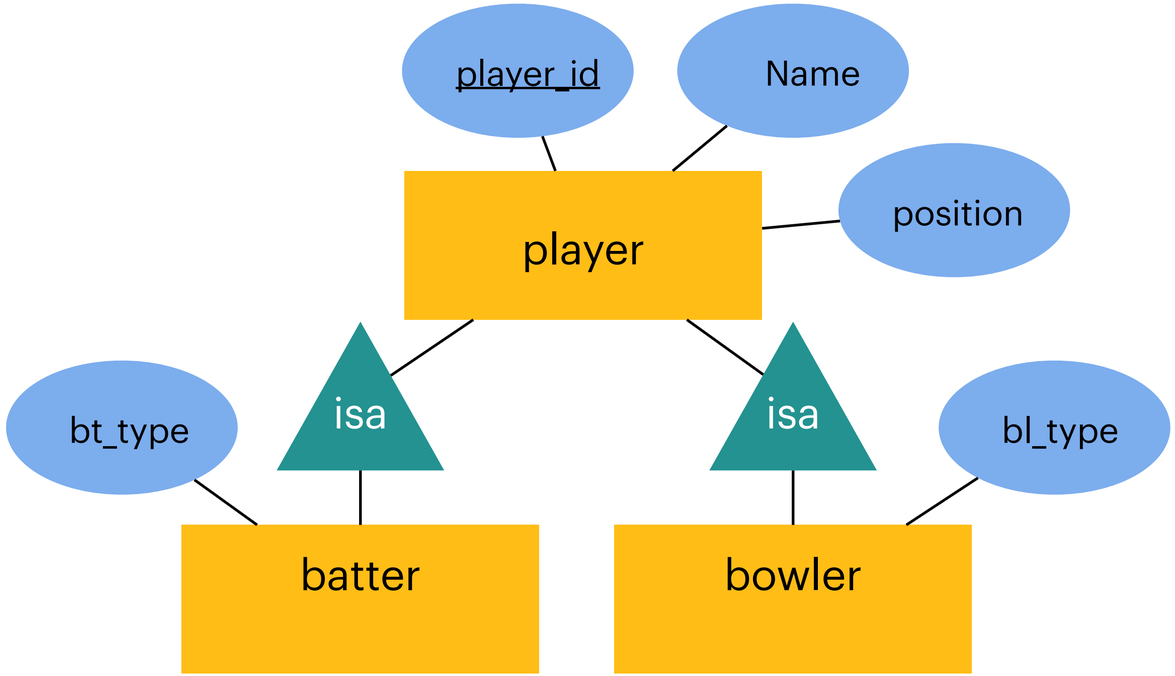
- [Player]: {[player_id, name, pos]}
- [Batter]: {[player_id, name, pos, bt_type]}
- [Bowler]: {[player_id, name, pos, bl_type]}



IsA Relationship

Option-4

- [Player]: {[player_id, name, pos]}
- [Batter]: {[player_id, bt_type]}
- [Bowler]: {[player_id, bl_type]}



Map to separate relations for the superclass and subclasses (option-4) or a single relation using a type attribute (option-1)

Relation vs Tables

Player =

(154, David Warner, 3, DC)
(123, MS Dhoni, 7, CSK)
(134, SK Yadav, 5, MI)
(131, Rohit Sharma, 1, MI)
(193, Virat Kohli, 3, RCB)

Player

PlayerId	Name	Position	PlaysFor
123	MS Dhoni	7	CSK
193	Virat Kohli	3	RCB
131	Rohit Sharma	1	MI
134	SK Yadav	5	MI
154	David Warner	3	DC

- Mathematical concept
- Tuples have no inherent order (unordered set)
- No duplicates (its a set!)
- Does not formally account for nulls

- Structure in a database to store data
- Rows have order (but does affect results)
- May allow duplicates (unless constrained)
- Often allow null values (missing data)

Summary

- Relational model fundamentals
- ER to relational mapping
  - Entities become relations with their attributes
  - Relationships are translated based on their cardinalities and participation
- Relational Model underpins modern databases, enabling efficient querying, data consistency, and scalability
- Next Lecture: Relational algebra — the query language of the relational model

Then vs Now: Same Fear New Domain?

- What was IBM afraid of in 1970?
  - Losing control over **how** things are done
  - Trusting the system to choose execution paths
  - Giving up hand-optimized procedures
- Replace
  - **IMS engineers** → ML engineers / prompt engineers
  - **Access paths** → training pipelines / inference strategies
  - **Query optimizer** → model reasoning / agent planner

Declarative AI?

Then vs Now: Same Fear New Domain?

Discussion Question

The relational model asked programmers to stop specifying access paths and trust the query optimizer.

Declarative AI systems ask developers to stop specifying pipelines and trust the model or agent.

Should we trust declarative systems with critical decisions—or is this a dangerous loss of control?

Declarative AI?